YOLOv11 Environment Construction

This chapter is applicable to self-construction of Jetson nano official image. If you use the YAHBOOM version of the image, this tutorial can be ignored.

1. Overview

YOLO11 is Ultralytics YOLO is the latest product in the real-time object detector series, redefining what is possible with state-of-the-art accuracy, speed, and efficiency. Building on the impressive progress of previous YOLO versions, YOLO11 has made significant improvements in architecture and training methods, making it a versatile choice for a wide range of computer vision tasks.

2. Environmental requirements

Due to limitations, only the older version of **JetPack**4.6.6 can be burned on **jetson nano**. The version used in our image is **JetPack**4.6.3. In this version, **CUDA** 10.2, **cuDNN** 8.2.1, **TensorRT** 8.2.1, and the system comes with **python** version 3.6.

However, the **python** version required to install Ultralytics is 3.8, so this article chooses to use **docker** to install.

3. Docker setup

3.1. Official installation of Docker

If Docker is not installed, you can use the script to install Docker in one click.

```
Domestic users may not be able to install Docker through official methods. We
recommend using Alibaba Cloud to install Docker or directly using our image.
```

- Update the local package list

```
sudo apt update
```

- Upgrade the installed package

```
sudo apt upgrade
```

- Download and run the script

Download the get-docker.sh file and save it in the current directory.

```
sudo apt install curl
curl -fsSL https://get.docker.com -o get-docker.sh
```

Use sudo privileges to run the get-docker.sh script file.

```
sudo sh get-docker.sh
```

3.2, Alibaba Cloud install Docker

If you cannot install it yourself, please use the image we provide.

- Update the local package list

```
sudo apt update
```

- Install the required software

```
sudo apt install apt-transport-https ca-certificates curl gnupg2 lsb-release
software-properties-common
```

- Add the GPG key of the software source

```
curl -fsSL https://mirrors.aliyun.com/docker-ce/linux/ubuntu/gpg | sudo gpg --
dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg
```

- Add Alibaba Cloud mirror software source

```
echo "deb [arch=arm64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
https://mirrors.aliyun.com/docker-ce/linux/ubuntu bionic stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null
```

- Install Docker

```
sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

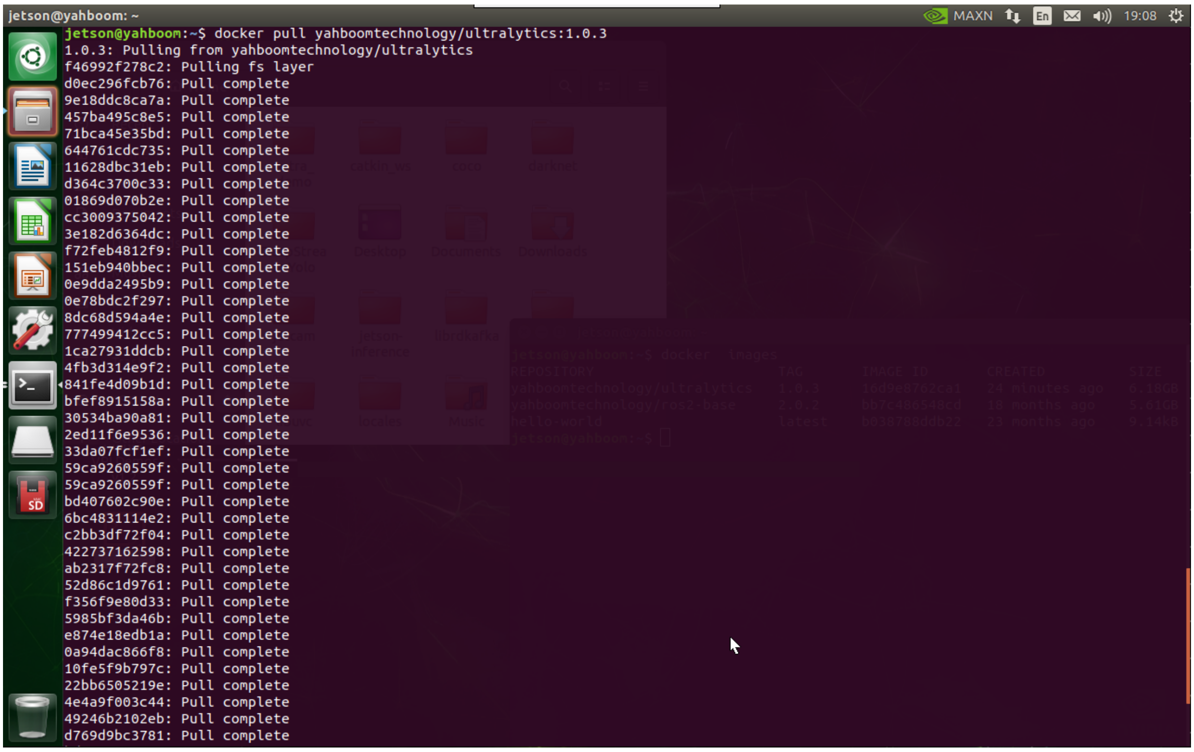
3.3. Add access rights

Add the current user's access rights to the Docker daemon: You can use Docker commands without using the sudo command

```
sudo usermod -aG docker $USER
newgrp docker
```

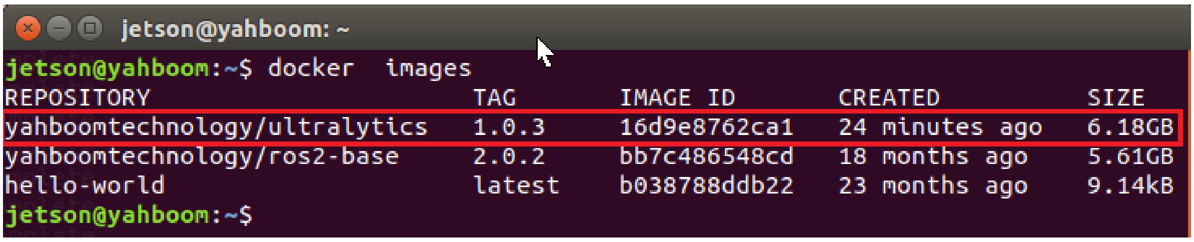
4. Install Ultralytics

```
docker pull yahboomtechnology/ultralytics:1.0.3
```



Query docker image

```
docker images
```



5. Edit the startup script

[yolov11_docker.sh] The contents of the script are as follows:

```
#!/bin/bash
xhost +

docker run -it \
--net=host \
--env="DISPLAY" \
--env="QT_X11_NO_MITSHM=1" \
-v /tmp/.X11-unix:/tmp/.X11-unix \
-v /home/jetson/temp:/ultralytics/ultralytics/temp \
--device=/dev/video0 \
-p 9090:9090 \
-p 8888:8888 \
yahboomtechnology/ultralytics:1.0.3 /bin/bash
```

Commented script description:

Note: When adding a host device to the container below, if the host is not connected to the device, you need to remove the corresponding addition operation to start the container

```
#!/bin/bash
xhost +

docker run -it \
--net=host \
--env="DISPLAY" \
--env="QT_X11_NO_MITSHM=1" \
-v /tmp/.X11-unix:/tmp/.X11-unix \
-v /home/jetson/temp:/ultralytics/ultralytics/temp \
--device=/dev/video0 \ # USB camera, add a host device to the container. If the camera is not connected, please remove this line. Video* needs to be modified according to the specific device port found in your system
-p 9090:9090 \
-p 8888:8888 \
yahboomtechnology/ultralytics:1.0.3 /bin/bash
```

6. Execute the script

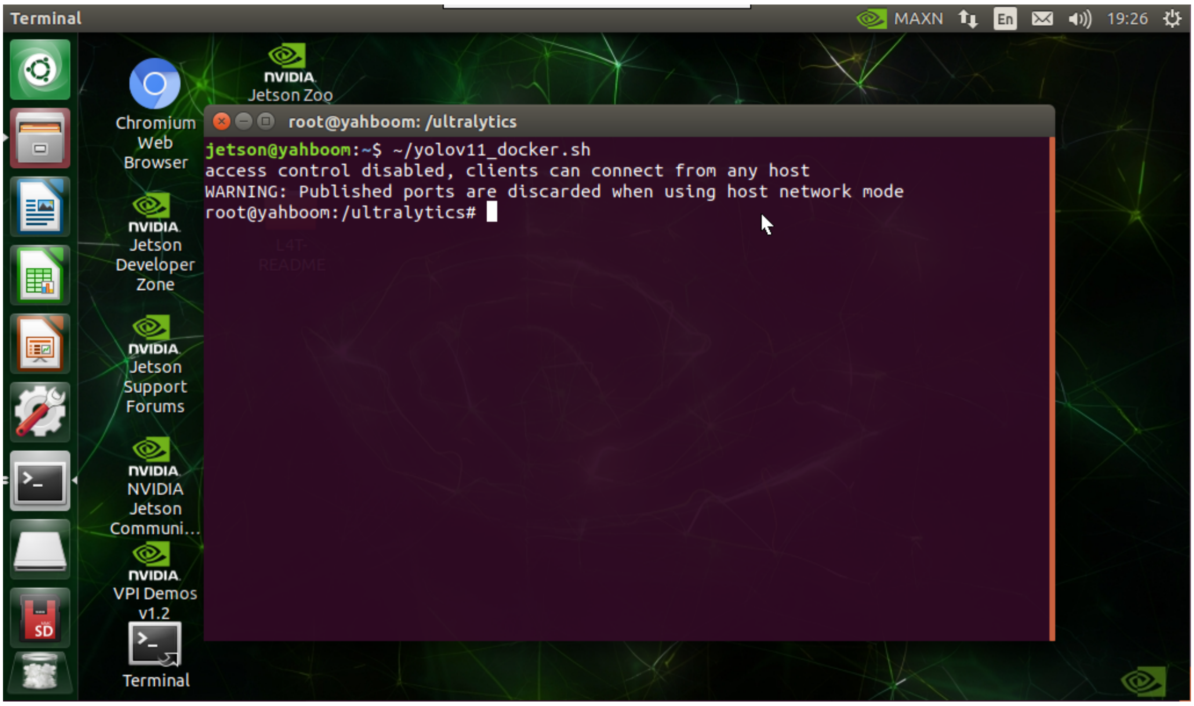
After editing the docker startup script, open the terminal on the docker host (on VNC or on the motherboard display)

Note: This must be executed on VNC or on the motherboard display, and cannot be executed in a terminal accessed remotely via ssh (such as a terminal accessed via MobaXterm), otherwise the GUI image may not be displayed in the container.

Run the previously created startup script in the VNC interface or on the screen (Note: Each execution of the script creates a new container from the image)

```
./yolov11_docker.sh
```

You can enter the container correctly, and run the real-time detection case to display the camera image.



References

https://docs.ultralytics.com/guides/nvidia-jetson/